

Ali Rashidi

Computer Vision & Deep Learning Engineer

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Profile Summary

Computer Vision and Deep Learning Engineer with hands-on experience designing, training, and deploying end-to-end CV pipelines across medical imaging, Earth Observation, and astronomical data. Skilled in CNNs, Transformers, YOLO, and U-Net architectures for object detection, semantic segmentation, image classification, and anomaly detection. Currently leading thermal-infrared imaging performance and Signal-to-Noise Ratio (SNR) modeling for the IGNIS CubeSat Earth-observation mission, with proven experience packaging deep learning models as Dockerized REST inference services on GCP.

Professional Experience

Computer Vision Engineer – Thermal Imaging Payload (IGNIS CubeSat)

University of Naples “Federico II” – IGNIS CubeSat Mission | Naples, Italy | 03/2025 – Present

- Developed physics-aware imaging performance and Signal-to-Noise Ratio (SNR) models for a thermal-infrared Earth Observation CubeSat payload, supporting feasibility of satellite-derived imagery.
- Built Python-based simulation pipelines (NumPy, SciPy, Matplotlib) to evaluate Field of View (FoV), spatial resolution, ground footprint, and radiometric signal under realistic orbital conditions.
- Applied time-series anomaly detection and trend analysis to mission simulation outputs, identifying non-optimal launch timing and recommending a ~5-month launch postponement to improve early-mission imaging quality.
- Translated mission analysis data into quantitative imaging performance predictions, enabling change-detection-style assessment of remote sensing observation feasibility.
- Supported payload design trade-offs and engineering decision-making in collaboration with multidisciplinary spacecraft teams; produced technical reports and stakeholder presentations.

Data Analyst

EZ-Tech | 01/2020 – 12/2023

- Built and maintained Power BI dashboards tracking key business KPIs, reducing manual reporting effort by ~40%.
- Wrote complex SQL queries to extract, clean, and analyze large datasets, supporting data-driven decisions across sales, and operations.
- Performed exploratory data analysis and statistical testing (A/B tests, regression, cohort analysis) using Python (Pandas, NumPy) and Excel, turning findings into actionable recommendations.
- Collaborated with cross-functional teams to define KPIs and automate recurring reports, improving data accuracy and shortening reporting cycle time by ~30%.

Data Analytics Intern

Faraz Omran Tadbir | 03/2019 – 09/2019

- Supported data analysis tasks by writing SQL queries and building Excel / Power BI reports, helping the team track KPIs and identify trends across business operations.

Education

M.Sc. in Data Science

University of Naples Federico II | Naples, Italy | 09/2023 – Present

Thesis: Signal-to-Noise Ratio characterization for thermal-infrared microbolometric imaging in CubeSat missions.

Relevant coursework: Machine Learning, Computer Vision, Statistical Learning, Data Mining, Big Data, Advanced Statistical Modeling.

B.Sc. in Computer Software Engineering | Islamic Azad University

Technical Skills

- **Computer Vision & Deep Learning:** Image classification, object detection (2D/3D), semantic segmentation, instance segmentation, anomaly detection, image preprocessing & augmentation, feature extraction, CNNs, Transformers, ViT, YOLO, U-Net, transfer learning, hyperparameter tuning, model evaluation (mAP, IoU, Dice, precision/recall).
- **Programming & Frameworks:** Python, C++, PyTorch, TensorFlow / Keras, OpenCV, NumPy, Pandas, SciPy, Scikit-learn, Matplotlib, Pillow, Jupyter Notebook.
- **MLOps & Deployment:** Docker, REST inference APIs, Google Cloud Platform (GCP), Google Cloud Run, CI/CD, Git, model versioning, model packaging, containerized deployment, performance optimization.
- **Imaging Systems & Remote Sensing:** Thermal-infrared imaging, radiometric modeling, Signal-to-Noise Ratio (SNR), Field of View (FoV), microbolometric sensors, Earth Observation, satellite imagery, change detection, medical image preprocessing (Hounsfield Units, multi-windowing), edge-constrained imaging systems.
- **Software Engineering:** Version control (Git), software testing, debugging, performance troubleshooting, code-review workflows, Linux (bash / csh), Arduino, Zemax OpticStudio.
- **Soft Skills:** Analytical problem-solving, engineering judgment, cross-disciplinary collaboration, technical communication & documentation, leadership & project coordination.

Selected Projects

AI-Assisted Intracranial Hemorrhage Segmentation from Brain CT *02/2026 – Present*

- Designed a modular medical-AI pipeline orchestrating Computer Vision, Machine Learning, LLM, and RAG components for hemorrhage detection, segmentation, and clinical decision support from 3D CT scans.
- Implemented end-to-end medical preprocessing preserving Hounsfield Units, applying multi-windowing into 3-channel representations, brain extraction (threshold + bounding box + crop), and 3D-to-2D slicing.
- Trained a U-Net segmentation model with combined Dice + BCEWithLogitsLoss, addressed severe class imbalance via balanced patch sampling (50% positive / 50% negative).
- Enforced scan-level data-leakage prevention and annotation quality control to ensure reliable clinical learning across heterogeneous CT datasets.

Brain Tumor Segmentation using YOLOv8 (Multimodal CT and MRI) *02/2026 – Present*

- Trained a YOLOv8-seg model with transfer learning on multimodal CT and MRI datasets in a GPU-accelerated environment, generating pixel-level tumor masks.
- Evaluated convergence and generalization with box mAP, mask mAP, precision, recall, loss curves, and confusion matrix analysis; ran systematic hyperparameter tuning and augmentation strategies for cross-modality robustness.

Morphological Classification of Extragalactic Radio Sources using CNN *08/2024 – 11/2024*

- Designed and trained a CNN classifier for radio galaxies (FRO, FRI, FRII) on grayscale astronomical imagery, achieving ~80–85% accuracy after class balancing and augmentation.
- Built an end-to-end CV pipeline (dataset preparation, preprocessing, training, evaluation) and analyzed class-wise performance differences driven by morphological complexity.
- Packaged the trained model as a Dockerized REST inference service and deployed it on Google Cloud Run, validating portability and inference stability through cloud-based testing.

Languages

English – C1 **Italian** – A2 **German** – A1 **Persian** – Native

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